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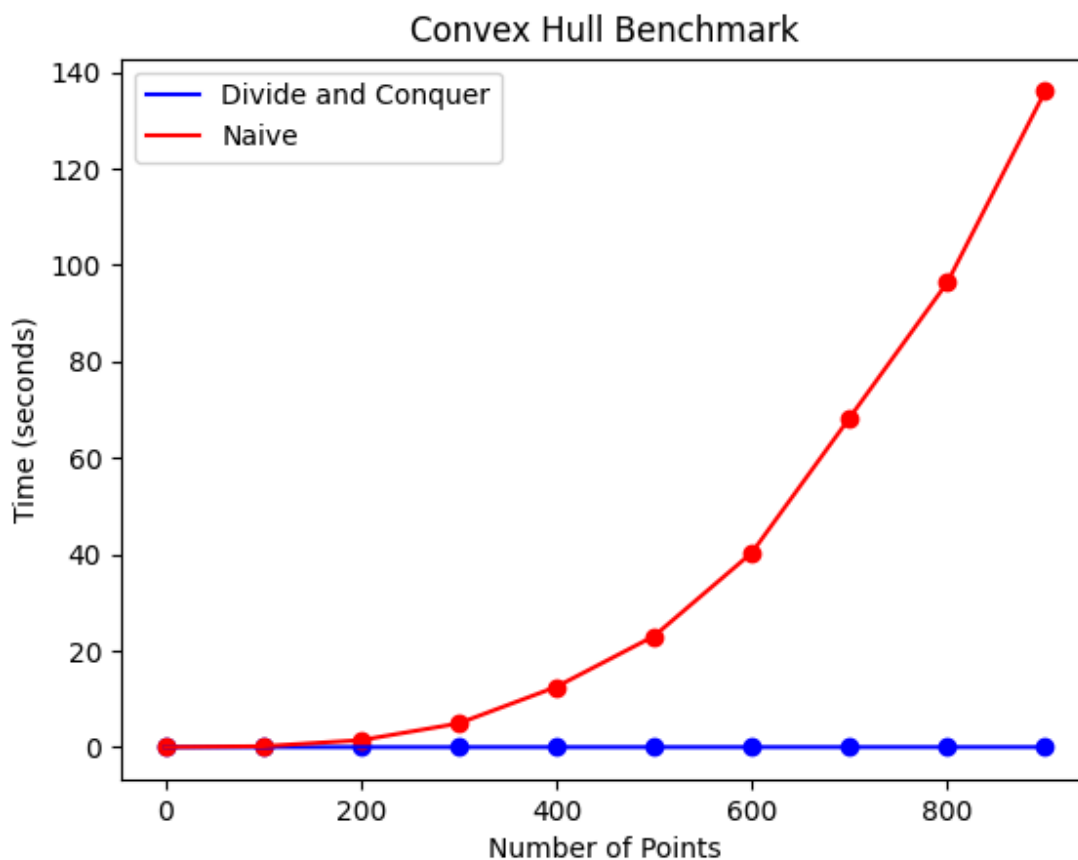
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CSC 440 - Design and Analysis of Algorithms

Assignment 2 - Convex Hull Algorithm

Benchmarking the Algorithms

The graph below was generated using the Python script provided with the assignment. This specific benchmark measured the execution time of both the divide and conquer and naive algorithms. We opted to run the benchmark on sample sizes between zero and one thousand at increments of one hundred. We believed these sizes best highlighted the differences between the two algorithms while still being a reasonable task for the naive algorithm.



Interpretation of the Algorithm

The results of our benchmarks support our expectations for the asymptotic complexity of this algorithm.

The results of the benchmarking did support our expectations that the base case algorithm runs in $O(n^3)$ time and the divide and conquer algorithm runs in $O(n \log n)$.

Do they support your expectation of the asymptotic complexity of this algorithm?

Why or why not?

What about the naive (base case) algorithm?